TERRENCE LEI

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EDUCATION

University of California - Santa Barbara **August 2023 - June 2027**
Bachelor of Science (BS), Mechanical Engineering
- Specialization in Dynamic Systems, Control, and Robotics
- Relevant Coursework: Controls, Mechatronics, Fluids, Thermodynamics, Machine Learning, Circuits, Vibrations

University of California - Santa Barbara **August 2023 - June 2027**
Bachelor of Science (BS), Statistics and Data Science

PROFESSIONAL EXPERIENCE

Genentech **September 2026 - Present**
Capstone Student *San Francisco, CA, USA*
- Developing a closed-loop robotic workcell that dispenses pharmaceutical powders with an xArm 6 to targets of 0.1–100 g at 0.01 g resolution, across free-flowing, cohesive and static-prone materials
- Solving the arm's kinematics and motion planning in ROS 2 Humble with MoveIt, coupling coarse collision-free moves with a fine controller that reads the analytical balance over serial and adapts spatula tilt and tapping to each powder
- Training dispensing policies with reinforcement learning in NVIDIA Isaac Sim and transferring them to hardware, with a vision-language model on workcell cameras to verify powder state, scoop fill and target completion

Applied Materials **June 2026 - September 2026**
Technical Project Management Intern *Santa Clara, CA, USA*
- Designed a patent pending motorized actuator mechanism capable of two distinct directions of movement from a single motor source while preserving manual release functionality and zero backlash
- Positioned as an upgrade with new capabilities including sensor data, automation solutions, and predictive insights for 1000+ tools worldwide

University of California - Santa Barbara **January 2026 - June 2026**
Undergraduate Research Assistant *Santa Barbara, CA, USA*
- Developing automated system to monitor workbench status, robotic movements and report results including a custom camera sensor suite
- Training a robotic arm for automated testing sequences, with the potential to support 50k material tests a day
- Designed data flow from raw test results managed with SQL to actionable insights

Northrop Grumman **June 2025 - August 2025**
Mechanical Engineering Intern *Goleta, CA, USA*
- Designed and conducted component and process experiments for assembly failure analysis, using regression and classification methods to identify the root causes and reduce production times and material costs
- Engineered a solar array test vehicle used to develop wiring techniques, optimizing assembly ergonomics to reduce lead times
- Worked on tolerance analysis for legacy flight components and released new 2D GD&T drawings

PROJECTS & OUTSIDE EXPERIENCE

Autonomous Shopping Cart **April 2026 - June 2026**
Undergraduate Student *Santa Barbara, CA, USA*
- Engineered an autonomous shopping cart as an integrated consumer product, translating shopper behavior into requirements for form factor, maneuverability, safety, controls, and an intuitive user experience
- Integrated a complex electromechanical system spanning power electronics, onboard compute, and machine vision
- Iteratively tested the interaction experience including following, obstacle avoidance, aisle navigation, target reacquisition, and visual feedback

Gaucho Rocket Project **September 2024 - June 2026**
Vice President *Santa Barbara, CA, USA*
- Performed FEA and structural optimization of flight mechanisms; iterating load paths and material selection to minimize mass while meeting stiffness, strength, and factor of safety requirements
- Designed for manufacturing across CNC machining, additive manufacturing, sheet metal, and composite processes
- Organized and led design reviews in partnership with faculty members and industry sponsors

SKILLS

Programming: C/C++, Java, Python, SQL, HTML/CSS, R, MATLAB, LaTex, VS Code, PyCharm, ROS, Pandas, Pytorch
Technical: SolidWorks, Fusion 360, Ansys CFX, MSC Adams, Creo, Windchill, Simulink, Excel, GD&T, LabView, NX, Siemens Teamcenter